Sahaj Bhadja

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Summary

Full-stack Software Engineer with 2+ years building **TypeScript**, **React**, and **Node.js** features over **PostgreSQL**, background job processing, and microservices, with automated testing, CI/CD releases, and rapid AI prototypes.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed normalized **PostgreSQL** schemas with indexing and data validation for 200+ users and 10K+ records, running SQL transformations across PostgreSQL and SQL Server that reduced query latency 35%.
- Restructured FastAPI **background workers** with explicit job-state transitions and failure isolation, processing 1K+ async jobs at 3x throughput, and shipped integrations through CI/CD-backed releases at 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built **React** components with hooks, state, and upload forms for question, rubric, and grading-history views, integrated end to end with REST APIs and serving 150+ instructors and 10K+ daily requests.
- Prototyped and shipped **AI** grading workflows across GPT-4o, Claude, and Qwen-VL with OCR fallbacks and batch processing, cutting feedback turnaround 60% while deploying through GitHub Actions to Vercel.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Built cross-platform data-sharing **microservices** with object-oriented Singleton and Factory modules, covered by 20+ automated unit and integration test suites run nightly in Jenkins across Windows, Linux, and macOS.
- **Troubleshot** and optimized API integrations with pagination, field filtering, indexing, and caching, improving response times 30% and cutting technical debt 20% through module refactoring in Agile sprints.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **TypeScript** platform with React and Node.js Express microservices for authentication, billing, and leads, persisting data in PostgreSQL and Azure SQL and syncing 2M+ records across locations.
- Wrote **Jest** unit tests and Selenium end-to-end tests gating Jenkins CI/CD releases with Docker packaging and staged deployments, reaching 95%+ successful build validation and halving deployment time.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built a **React and TypeScript** web dashboard and Flask APIs for bookings, pricing, and coupons with Firebase real-time status updates, keeping responses under 200ms across 50K+ records.
- Automated refunds and disputes through idempotent **payment webhooks**, cutting manual handling about 70%, and presented the end-to-end architecture to the client after mapping 12+ backend integration points.

Skills

Languages: TypeScript, JavaScript, Python, SQL, Java, HTML, CSS
Frontend & AI: React, Next.js, TailwindCSS, React Native, WebSockets, LLM APIs, Prompt Engineering
Backend: Node.js, Express.js, REST APIs, Microservices, FastAPI, Flask, Async Processing, Background Jobs
Data: PostgreSQL, SQL Server, Azure SQL, MySQL, Schema Design, Indexing, Data Validation
Testing & Delivery: Jest, Selenium, PyTest, Unit Testing, Integration Testing, CI/CD, Jenkins, GitHub Actions, Docker, Git, AWS

Projects

Employee Leave Request Platform | *TypeScript, Next.js, Node.js, GraphQL, PostgreSQL, Jest*

- Designed a **Next.js** and Node.js **GraphQL** app for leave requests and manager approvals, with PostgreSQL migrations, queued background jobs for notifications and policy checks, and Jest and React Testing Library tests.

HR Policy Assistant Prototype | *TypeScript, Next.js, LLM APIs, Embeddings, PostgreSQL*

- Prototyped a **retrieval-based** assistant that answers employee questions about benefits and handbook policies with citations to source documents, plus an evaluation set to catch wrong or **unsupported answers**.

FinNexus Live | *Next.js, TypeScript, TailwindCSS, MongoDB, Kubernetes, WebSockets*

- Built a **Next.js** and TypeScript fintech app with TailwindCSS, Plaid bank linking, real-time WebSocket updates, and transaction filtering, supporting 50,000+ users and 1M+ transactions with CI/CD on Kubernetes.